CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 2: Divide and Conquer Algorithms Analysis

Course: Design Analysis and Algorithms [23CS3502]
Category: Assignments | Assignment #2 | Academic Year: 2026-27

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1. Overview & Objective:

Divide and conquer algorithmic paradigm, divide overhead, and recurrence solutions.

2. Assignment Problems & Instructions:

1. Derive the exact recurrence and recursion tree for Merge Sort: T n = 2T n/2 + O n .
2. Analyze randomized QuickSort expected time complexity of O n log n .
3. Derive Strassen matrix multiplication recurrence T n = 7T n/2 + O n^2 .
4. Implement and analyze the Closest Pair of Points 2D geometric algorithm.

Submission Guidelines & Verification:

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